

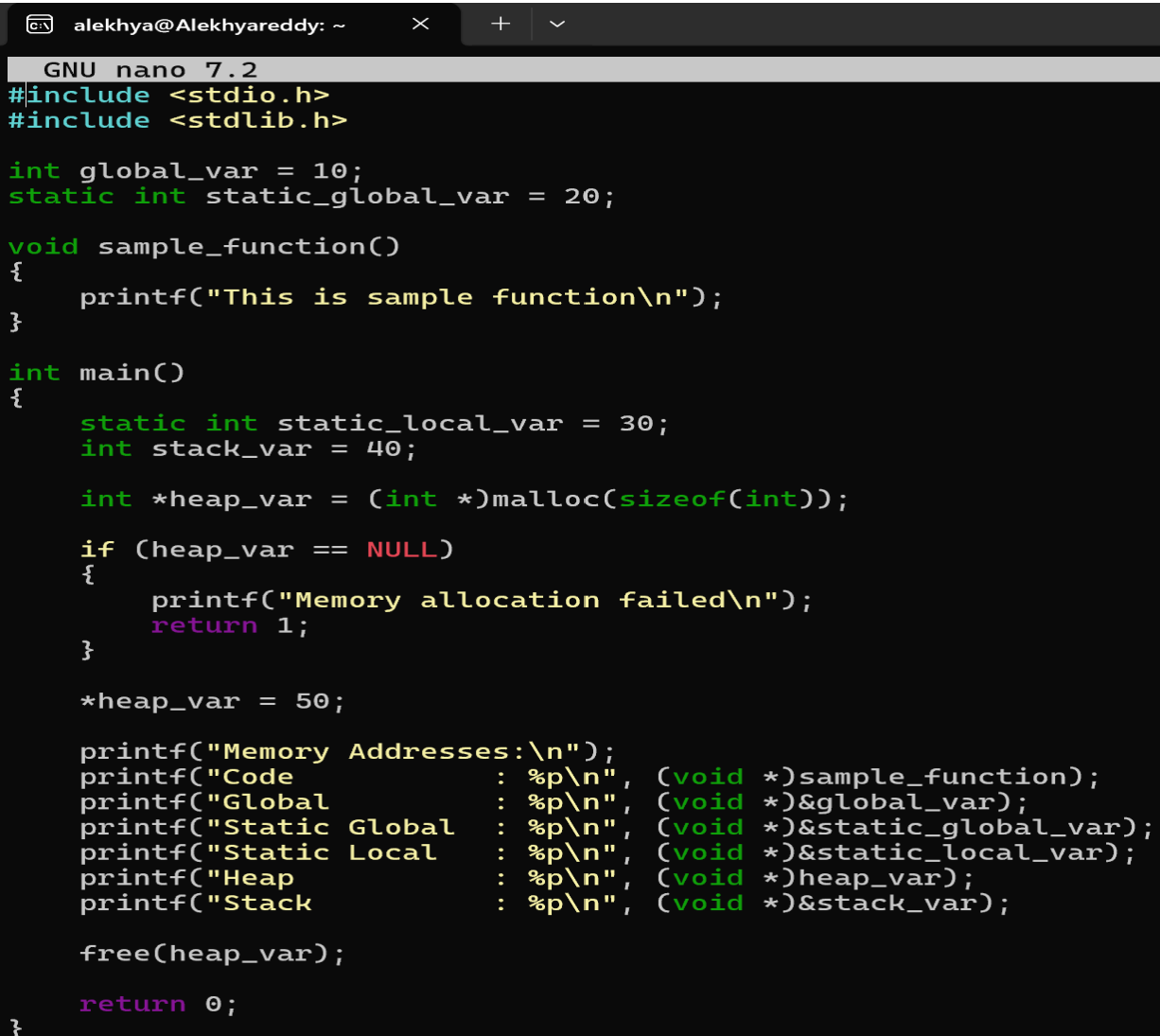
WEEK-7 PRACTICAL

1 . Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Aim :

To write a C program that displays the memory addresses of variables belonging to the code, global, static, heap, and stack segments and understand the Linux process address-space layout.

Program :



```
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

int global_var = 10;
static int static_global_var = 20;

void sample_function()
{
    printf("This is sample function\n");
}

int main()
{
    static int static_local_var = 30;
    int stack_var = 40;

    int *heap_var = (int *)malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    *heap_var = 50;

    printf("Memory Addresses:\n");
    printf("Code          : %p\n", (void *)sample_function);
    printf("Global          : %p\n", (void *)&global_var);
    printf("Static Global    : %p\n", (void *)&static_global_var);
    printf("Static Local     : %p\n", (void *)&static_local_var);
    printf("Heap             : %p\n", (void *)heap_var);
    printf("Stack            : %p\n", (void *)&stack_var);

    free(heap_var);

    return 0;
}
```

Output :

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ nano memory.c  
alekhya@Alekhyareddy:~$ gcc memory.c -o memory  
alekhya@Alekhyareddy:~$ ./memory  
Memory Addresses:  
Code      : 0x5957ad5171c9  
Global    : 0x5957ad51a010  
Static Global : 0x5957ad51a014  
Static Local  : 0x5957ad51a018  
Heap       : 0x5957c2a5e2a0  
Stack      : 0x7fff650476dc  
alekhya@Alekhyareddy:~$ ./memory  
Memory Addresses:  
Code      : 0x5eeb41b391c9  
Global    : 0x5eeb41b3c010  
Static Global : 0x5eeb41b3c014  
Static Local  : 0x5eeb41b3c018  
Heap       : 0x5eeb4f0132a0  
Stack      : 0x7ffc97956f5c  
alekhya@Alekhyareddy:~$ |
```

2 . Make use of the `/proc/maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Aim :

To study the memory organization of a running Linux process using the `/proc/<PID>/maps` file and memory-analysis commands, and understand virtual memory mappings.

Program :

```
alekhya@Alekhyareddy: ~  
GNU nano 7.2  
#include <stdio.h>  
#include <unistd.h>  
  
int global_var = 10;  
  
int main()  
{  
    int local_var = 20;  
  
    printf("Process is running...\n");  
    printf("PID: %d\n", getpid());  
  
    printf("Address of global variable: %p\n", (void *)&global_var);  
    printf("Address of local variable : %p\n", (void *)&local_var);  
  
    printf("\nUse the following commands in another terminal:\n");  
    printf("cat /proc/%d/maps\n", getpid());  
    printf("cat /proc/%d/status\n", getpid());  
    printf("cat /proc/%d/smaps\n", getpid());  
    printf("pmap %d\n", getpid());  
  
    printf("\nProcess will continue running...\n");  
  
    while (1)  
    {  
        sleep(10);  
    }  
  
    return 0;  
}
```

Output :

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ pidof sleep  
639  
alekhya@Alekhyareddy:~$ cat /proc/639/maps  
5aa20b502000-5aa20b504000 r--p 00000000 08:30 49348 /usr/bin/sleep  
5aa20b504000-5aa20b508000 r-xp 00002000 08:30 49348 /usr/bin/sleep  
5aa20b508000-5aa20b509000 r--p 00006000 08:30 49348 /usr/bin/sleep  
5aa20b509000-5aa20b50a000 r--p 00007000 08:30 49348 /usr/bin/sleep  
5aa20b50a000-5aa20b50b000 rw-p 00008000 08:30 49348 /usr/bin/sleep  
5aa210cdb000-5aa210cfc000 rw-p 00000000 00:00 0 [heap]  
7c1170c00000-7c1170c28000 r--p 00000000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
7c1170c28000-7c1170db0000 r-xp 00028000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
7c1170db0000-7c1170dff000 r--p 001b0000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
7c1170dff000-7c1170ec3000 r--p 001fe000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
7c1170ec3000-7c1170ec5000 rw-p 00202000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
7c1170ec5000-7c1170e12000 rw-p 00000000 00:00 0  
7c1170e66000-7c1170ebf000 r--p 00000000 08:30 12181 /usr/lib/locale/C.utf8/LC_CTYPE  
7c1170ebf000-7c1170ec0000 r--p 00000000 08:30 12187 /usr/lib/locale/C.utf8/LC_NUMERIC  
7c1170ec0000-7c1170ec1000 r--p 00000000 08:30 12190 /usr/lib/locale/C.utf8/LC_TIME  
7c1170ec1000-7c1170ec2000 r--p 00000000 08:30 12180 /usr/lib/locale/C.utf8/LC_COLLATE  
7c1170ec2000-7c1170ec3000 r--p 00000000 08:30 12185 /usr/lib/locale/C.utf8/LC_MONETARY  
7c1170ec3000-7c1170ec4000 r--p 00000000 08:30 12184 /usr/lib/locale/C.utf8/LC_MESSAGES/SYS_LC_MESSAGES  
7c1170ec4000-7c1170ec5000 r--p 00000000 08:30 12188 /usr/lib/locale/C.utf8/LC_PAPER  
7c1170ec5000-7c1170ec6000 r--p 00000000 08:30 12186 /usr/lib/locale/C.utf8/LC_NAME  
7c1170ec6000-7c1170ec7000 r--p 00000000 08:30 12179 /usr/lib/locale/C.utf8/LC_ADDRESS  
7c1170ec7000-7c1170ec8000 r--p 00000000 08:30 12189 /usr/lib/locale/C.utf8/LC_TELEPHONE  
7c1170ec8000-7c1170ec9000 r--p 00000000 08:30 12183 /usr/lib/locale/C.utf8/LC_MEASUREMENT  
7c1170ec9000-7c1170ecc000 rw-p 00000000 00:00 0  
7c1170ecc000-7c1170ed3000 r--s 00000000 08:30 49751 /usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache  
7c1170ed3000-7c1170ed4000 r--p 00000000 08:30 12182 /usr/lib/locale/C.utf8/LC_IDENTIFICATION  
7c1170ed4000-7c1170ed6000 rw-p 00000000 00:00 0  
7c1170ed6000-7c1170eda000 r--p 00000000 00:00 0 [vvar]  
7c1170eda000-7c1170edc000 r--p 00000000 00:00 0 [vvar_vclock]  
7c1170edc000-7c1170ede000 r-xp 00000000 00:00 0 [vdso]  
7c1170ede000-7c1170edf000 r--p 00000000 08:30 49759 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7c1170edf000-7c1170f0a000 r-xp 00001000 08:30 49759 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7c1170f0a000-7c1170f14000 r--p 0002c000 08:30 49759 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7c1170f14000-7c1170f16000 r--p 00036000 08:30 49759 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7c1170f16000-7c1170f18000 rw-p 00038000 08:30 49759 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2  
7ffefe5a0000-7ffefe5c1000 rw-p 00000000 00:00 0 [stack]
```

```
alekhya@Alekhyareddy: ~  
alekhya@Alekhyareddy:~$ cat /proc/639/status  
Name: sleep  
Umask: 0022  
State: S (sleeping)  
Tgid: 639  
Ngid: 0  
Pid: 639  
PPid: 618  
TracerPid: 0  
Uid: 1000 1000 1000 1000  
Gid: 1000 1000 1000 1000  
FDSize: 256  
Groups: 27 100 1000  
NSTgid: 639  
NSpid: 639  
NSpgid: 639  
NSSid: 618  
Kthreadd: 0  
VmPeak: 3132 kB  
VmSize: 3132 kB  
VmLck: 0 kB  
VmPin: 0 kB  
VmHWM: 1904 kB  
VmRSS: 1904 kB  
RssAnon: 112 kB  
RssFile: 1792 kB  
RssShmem: 0 kB  
VmData: 224 kB  
VmStk: 132 kB  
VmExe: 16 kB  
VmLib: 1748 kB  
VmPTE: 44 kB  
VmSwap: 0 kB  
HugetlbPages: 0 kB  
CoreDumping: 0  
THP_enabled: 1  
untag_mask: 0xfffffffffffffffffffff  
Threads: 1  
SigQ: 1/62931  
SigPnd: 00000000000000000000000000000000  
ShdPnd: 00000000000000000000000000000000  
SigBlk: 00000000000000000000000000000000  
SigIgn: 00000000000000000000000000000000
```



```
alekhya@AlekhyaReddy: ~  
AnonHugePages:      0 kB  
ShmemPmdMapped:    0 kB  
FilePmdMapped:     0 kB  
Shared_Hugetlb:    0 kB  
Private_Hugetlb:   0 kB  
Swap:              0 kB  
SwapPss:           0 kB  
Locked:            0 kB  
THPEligible:       0  
VmFlags: rd mr mw me sd  
5aa20b509000-5aa20b50a000 r--p 00007000 08:30 49348 /usr/bin/sleep  
Size:              4 kB  
KernelPageSize:    4 kB  
MMUPageSize:       4 kB  
Rss:               4 kB  
Pss:               4 kB  
Pss_Dirty:         4 kB  
Shared_Clean:      0 kB  
Shared_Dirty:      0 kB  
Private_Clean:     0 kB  
Private_Dirty:     4 kB  
Referenced:        4 kB  
Anonymous:         4 kB  
KSM:               0 kB  
LazyFree:          0 kB  
AnonHugePages:     0 kB  
ShmemPmdMapped:    0 kB  
FilePmdMapped:     0 kB  
Shared_Hugetlb:    0 kB  
Private_Hugetlb:   0 kB  
Swap:              0 kB  
SwapPss:           0 kB  
Locked:            0 kB  
THPEligible:       0  
VmFlags: rd mr mw me ac sd  
5aa20b50a000-5aa20b50b000 rw-p 00008000 08:30 49348 /usr/bin/sleep  
Size:              4 kB  
KernelPageSize:    4 kB  
MMUPageSize:       4 kB  
Rss:               4 kB  
Pss:               4 kB  
Pss_Dirty:         4 kB  
Shared_Clean:      0 kB
```

```

alekhy@AlekhyaReddy: ~
Shared_Dirty:      0 kB
Private_Clean:     0 kB
Private_Dirty:     4 kB
Referenced:        4 kB
Anonymous:         4 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:     0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPeligible:       0
VmFlags: rd wr mr mw me ac sd
5aa210cdb000-5aa210cfc000 rw-p 00000000 00:00 0      [heap]
Size:              132 kB
KernelPageSize:    4 kB
MMUPageSize:        4 kB
Rss:               20 kB
Pss:               20 kB
Pss_Dirty:         20 kB
Shared_Clean:       0 kB
Shared_Dirty:       0 kB
Private_Clean:      0 kB
Private_Dirty:      20 kB
Referenced:         20 kB
Anonymous:          20 kB
KSM:               0 kB
LazyFree:          0 kB
AnonHugePages:     0 kB
ShmemPmdMapped:    0 kB
FilePmdMapped:     0 kB
Shared_Hugetlb:    0 kB
Private_Hugetlb:   0 kB
Swap:              0 kB
SwapPss:           0 kB
Locked:            0 kB
THPeligible:       0
VmFlags: rd wr mr mw me ac sd
7c1170c00000-7c1170c28000 r--p 00000000 08:30 49762  /usr/lib/x86_64-linux-gnu/libc.so.6

```

```
alekhya@Alekhyareddy: ~  
Size: 160 kB  
KernelPageSize: 4 kB  
MMUPageSize: 4 kB  
Rss: 160 kB  
Pss: 9 kB  
Pss_Dirty: 0 kB  
Shared_Clean: 160 kB  
Shared_Dirty: 0 kB  
Private_Clean: 0 kB  
Private_Dirty: 0 kB  
Referenced: 160 kB  
Anonymous: 0 kB  
KSM: 0 kB  
LazyFree: 0 kB  
AnonHugePages: 0 kB  
ShmemPmdMapped: 0 kB  
FilePmdMapped: 0 kB  
Shared_Hugetlb: 0 kB  
Private_Hugetlb: 0 kB  
Swap: 0 kB  
SwapPss: 0 kB  
Locked: 0 kB  
THPeligible: 0  
VmFlags: rd mr mw me sd  
7c1170c28000-7c1170db0000 r-xp 00028000 08:30 49762 /usr/lib/x86_64-linux-gnu/libc.so.6  
Size: 1568 kB  
KernelPageSize: 4 kB  
MMUPageSize: 4 kB  
Rss: 992 kB  
Pss: 48 kB  
Pss_Dirty: 0 kB  
Shared_Clean: 992 kB  
Shared_Dirty: 0 kB  
Private_Clean: 0 kB  
Private_Dirty: 0 kB  
Referenced: 992 kB  
Anonymous: 0 kB  
KSM: 0 kB  
LazyFree: 0 kB  
AnonHugePages: 0 kB  
ShmemPmdMapped: 0 kB  
FilePmdMapped: 0 kB  
Shared_Hugetlb: 0 kB
```


alekhya@AlekhyaReddy: ~

7c1170ed4000-7c1170ed6000 rw-p 00000000 00:00 0

Size: 8 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 4 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 0 kB
Private_Dirty: 4 kB
Referenced: 4 kB
Anonymous: 4 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
Shared_Hugetlb: 0 kB
Private_Hugetlb: 0 kB
Swap: 0 kB
SwapPss: 0 kB
Locked: 0 kB
THPEligible: 0

VmFlags: rd wr mr mw me ac sd

7c1170ed6000-7c1170eda000 r--p 00000000 00:00 0

[vvar]

Size: 16 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 0 kB
Pss: 0 kB
Pss_Dirty: 0 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 0 kB
Private_Dirty: 0 kB
Referenced: 0 kB
Anonymous: 0 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB

alekhya@Alekhyareddy: ~

```
SwapPss:          0 kB
Locked:           0 kB
THPEligible:      0
VmFlags: rd wr mr mw me gd ac
alekhya@Alekhyareddy:~$ pmap 639
639:   sleep 300
00005aa20b502000      8K r---- sleep
00005aa20b504000     16K r-x-- sleep
00005aa20b508000      4K r---- sleep
00005aa20b509000      4K r---- sleep
00005aa20b50a000      4K rw--- sleep
00005aa210cdb000    132K rw--- [ anon ]
00007c1170c00000    160K r---- libc.so.6
00007c1170c28000   1568K r-x-- libc.so.6
00007c1170db0000    316K r---- libc.so.6
00007c1170dff000     16K r---- libc.so.6
00007c1170e03000      8K rw--- libc.so.6
00007c1170e05000     52K rw--- [ anon ]
00007c1170e66000   356K r---- LC_CTYPE
00007c1170ebf000      4K r---- LC_NUMERIC
00007c1170ec0000      4K r---- LC_TIME
00007c1170ec1000      4K r---- LC_COLLATE
00007c1170ec2000      4K r---- LC_MONETARY
00007c1170ec3000      4K r---- SYS_LC_MESSAGES
00007c1170ec4000      4K r---- LC_PAPER
00007c1170ec5000      4K r---- LC_NAME
00007c1170ec6000      4K r---- LC_ADDRESS
00007c1170ec7000      4K r---- LC_TELEPHONE
00007c1170ec8000      4K r---- LC_MEASUREMENT
00007c1170ec9000     12K rw--- [ anon ]
00007c1170ecc000     28K r--s- gconv-modules.cache
00007c1170ed3000      4K r---- LC_IDENTIFICATION
00007c1170ed4000      8K rw--- [ anon ]
00007c1170ed6000     16K r---- [ anon ]
00007c1170eda000      8K r---- [ anon ]
00007c1170edc000      8K r-x-- [ anon ]
00007c1170ede000      4K r---- ld-linux-x86-64.so.2
00007c1170edf000   172K r-x-- ld-linux-x86-64.so.2
00007c1170f0a000     40K r---- ld-linux-x86-64.so.2
00007c1170f14000      8K r---- ld-linux-x86-64.so.2
00007c1170f16000      8K rw--- ld-linux-x86-64.so.2
00007ffefe5a0000   132K rw--- [ stack ]
total                3132K
```